

# WASP-50b

R-1152

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-50\_191016 · created 2026-07-31T01:22:40+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2458772.817 +/- 0.019 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.096 +/- 0.043               |
| Transit depth (Rp/Rs)^2                           | 0.92 +/- 0.83 [%]             |
| Orbital Inclination (inc)                         | 83.77 +/- 2.94                |
| Airmass coefficient 1 (a1)                        | 1.182 +/- 0.016               |
| Airmass coefficient 2 (a2)                        | -0.119 +/- 0.025              |
| Scatter in the residuals of the lightcurve fit is | 1.39 %                        |
| Best Comparison Star                              | None                          |
| Optimal Aperture                                  | 10.98                         |
| Optimal Annulus                                   | 25.83                         |
| Transit Duration (day)                            | 0.047 +/- 0.03                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.096 ± 0.043 vs 0.1404 (deviation 0.94σ) |
| Duration fit vs published | 0.047 d vs 0.0754 d (deviation 0.87σ)     |
| Mid-time O–C              | 45.5 min                                  |
| Depth SNR                 | 2.0                                       |
| Residual scatter          | 1.39 %                                    |

### Light curve

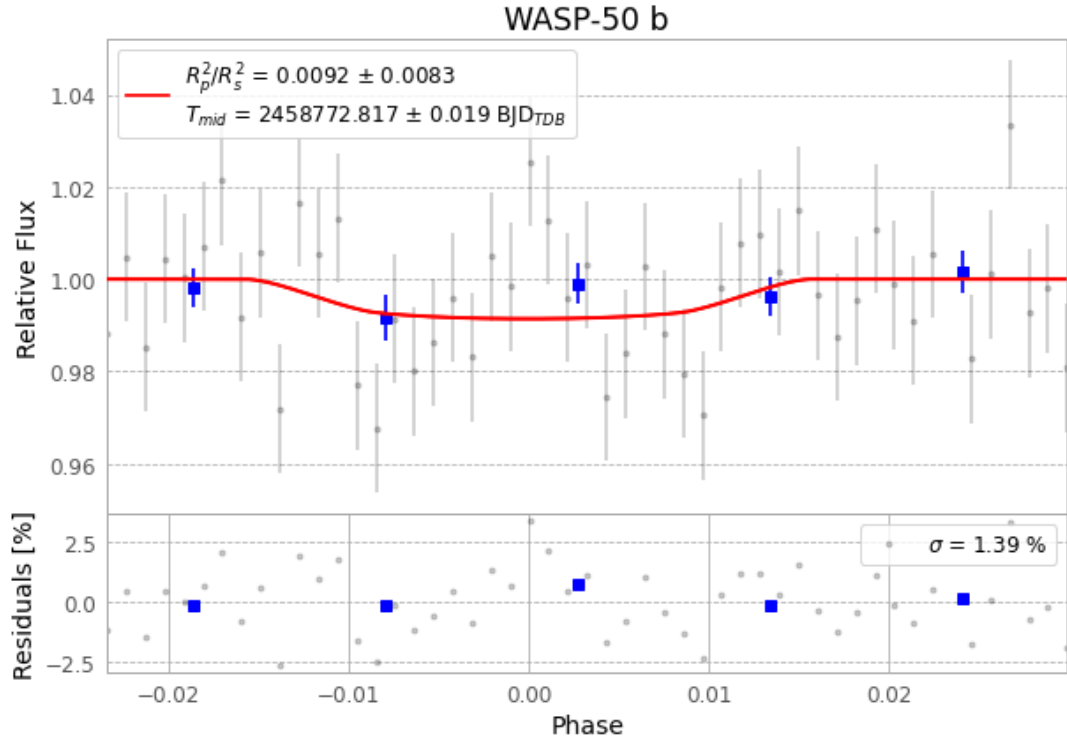


Figure 1 — FinalLightCurve\_WASP-50 b\_2019-10-15.png (SHA-256 07139a2acd70a972...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [159, 311], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 59021eb04128a212...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

51 FITS frames (33 MB), WASP-50191016062112.FITS → WASP-50191016085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-191016.log (2697576b0ce2...), FinalLightCurve\_WASP-50 b\_2019-10-15.png (07139a2acd70...), FinalLightCurve\_WASP-50 b\_2019-10-15.csv (94c28d47c3ef...)

## Compute resources

Wall time 225.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:22Z.